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A Project Report

On

Project Title: Smart Multi-Band Wireless Signal Blocking System

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(DST-FIST Sponsored Department)

DECLARATION

The Project Report entitled **"Smart Multi-Band Wireless Signal Blocking System"** is a record of Bonafide work of **2300049030-R. Komal Naidu, 2300049033- A. Adhithya, 2300049045- G. Abhiram Charan, 2300049048- G. Dakshinya Sneha** submitted in partial fulfilment for the award of B.Tech in Electronics and Communication Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the Project Report entitled "**SMART MULTI-BAND WIRELESS SIGNAL BLOCKING SYSTEM**" is being submitted **2300049030- R. Komal Naidu, 2300049033- A. Adhitya, 2300049045 – G. Abhiram Charan, 2300049048- G. Dakshinya Sneha** submitted in partial fulfilment for the award of B.Tech in Electronics and Communication Engineering to the K L University is a record of Bonafide work carried out under our guidance and supervision.

The results embodied in this report have not been copied from any other departments/ University/Institute.

Signature of Supervisor

Signature of the HOD

Signature of the External Examiner

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It is great pleasure for me to express my gratitude to our honourable President **Sri. Koneru Satyanarayana**, for giving the opportunity and platform with facilities in accomplishing the project report.

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ABSTRACT

The Smart Multi-Band Wireless Signal Blocking System is designed to prevent unauthorized wireless communication by blocking multiple frequency bands such as GSM, Wi-Fi, Bluetooth, and GPS. The system operates by generating interference signals in specific frequency ranges, thereby disrupting communication between transmitting and receiving devices.

This project focuses on developing a compact, efficient, and intelligent signal-blocking solution capable of selectively targeting multiple communication bands. The system can be integrated with a microcontroller-based control unit to enable selective band activation, power control, and smart operation. It is especially useful in secure environments where unauthorized wireless communication must be prevented to ensure safety, privacy, and discipline.

Literature Survey

1. Introduction to RF Monitoring and Detection

Wireless communication systems operate through radio frequency (RF) waves. Each technology uses specific frequency bands—for example, GSM operates around 900–1800 MHz, Wi-Fi around 2.4 GHz and 5 GHz, Bluetooth around 2.4 GHz, and GPS at 1.575 GHz. Detecting and analysing RF signals has been a key focus in communication research for ensuring secure and interference-free environments.

Several research works have focused on RF spectrum analysis, signal detection, and monitoring. Tools like spectrum analyser's and software-defined radios (SDRs) are widely used for RF surveillance, but they are often expensive and complex.

2. Wireless Signal Detection Techniques

2.1. Traditional Spectrum Analysers

Traditional spectrum analysers are used to visualize the frequency spectrum and observe active signals. They are widely used in RF labs to detect signal strength, modulation type, and signal presence. However, they are bulky, expensive, and not suited for embedded low-cost applications.

Limitation: Not suitable for portable or embedded systems in real-time monitoring.

2.2. Software Defined Radio (SDR)

SDRs like RTL-SDR, HackRF, and USRP can be programmed to receive a wide range of frequencies. They offer flexibility and are used in advanced RF research projects. SDRs can detect and demodulate signals and perform advanced spectrum analysis.

Advantage: Flexible frequency scanning and detailed spectrum visualization. *Disadvantage:* Requires powerful processing hardware and skilled programming.

2.3. RF Power Detection Modules

Low-cost RF detector modules, such as RF envelope detectors and power detectors, convert RF energy into voltage levels that approximate signal strength. These modules are widely used in embedded systems to estimate signal levels in a frequency band.

For example:

- **AD8318** — Wideband RF power detector
- **RF3220 / MAX2016** — RF power detectors for specific frequency ranges

Researchers have used these to sense wireless activity and estimate network presence.

Advantage: Simple, low-cost, suitable for microcontrollers
Limitation: Provides relative strength only; not full spectrum analysis.

3. Monitoring Wireless Networks in Restricted Zones

3.1. Monitoring GSM Signals

Studies have explored using mobile signal detectors for security enforcement. Academic papers show systems that detect active GSM communication to deter unauthorized use in secure zones. These systems often use GSM signal strength indicators and combine them with logging mechanisms.

Key Finding: Simple detection modules can be effective for indication, though not for blocking.

3.2. Wi-Fi Activity Detection

Wi-Fi signal detection has also been studied using both hardware modules and SDRs. Detecting 2.4 GHz Wi-Fi energy is simpler because many affordable RF power detectors are optimized in this range.

Key Finding: Wi-Fi detection is feasible with low-cost hardware, suitable for embedded monitoring.

3.3. Bluetooth and GPS Detection

Bluetooth and GPS signals have distinct RF characteristics. Bluetooth uses frequency-hopping spread spectrum (FHSS) in 2.4 GHz, while GPS uses a narrow 1575 MHz band.

Embedded systems designed for RF presence detection can detect energy peaks in these bands, though identifying specific protocols requires more advanced hardware.

4. Embedded Wireless Monitoring Systems

4.1. Microcontroller-Based RF Monitoring

Several research projects have used microcontrollers like Arduino, STM32, and ESP32 for RF monitoring applications. These designs interface RF detector modules, display units, and alert outputs to create autonomous monitoring devices.

Example:
A typical low-cost system uses a microcontroller with RF sensor inputs and displays measured signal levels on an LCD/OLED.

4.2. IoT-Based Wireless Monitoring

Some modern approaches integrate RF monitoring with IoT platforms. These systems upload signal strength data to cloud dashboards, enabling remote monitoring and analytics through web applications or mobile apps.

Finding: IoT integration improves system usability but increases complexity.

5. Gap Analysis

Feature	**Spectrum Analyzer**	**SDR**	**Low-Cost RF Monitoring**
Portability	✕	✕	✔
Cost	High	Medium-High	Low
Real-Time Embedded	✕	Requires Host	✔
Band Flexibility	High	Very High	Limited
Legal Compliance	✔	✔	✔

Traditional tools either cost too much or are not suitable for low-cost embedded monitoring. Low-cost RF detectors and microcontrollers strike a balance, allowing real-time signal monitoring in restricted environments.

6. Conclusion of Literature Survey

The literature shows that:

- RF spectrum analysis is critical for security applications.
- Traditional methods (spectrum analyser's & SDR) are powerful but expensive.
- Embedded monitoring using microcontrollers and RF power detectors is an effective, low-cost solution for wireless presence detection.
- There is limited research on multi-band embedded RF monitoring with alert systems, which is the focus of this project.

This project builds on previous research by combining multi-band RF detection with real-time display and alert mechanisms in a compact embedded system.

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Introduction

In today's modern world, wireless communication technologies such as GSM, Wi-Fi, Bluetooth, and GPS have become an integral part of daily life. These technologies operate on different radio frequency (RF) bands and enable seamless communication and data exchange. While they provide convenience and connectivity, they also create security concerns in sensitive or restricted areas such as examination halls, research laboratories, defence facilities, corporate meeting rooms, and confidential offices.

Unauthorized use of wireless devices in such environments may lead to data leakage, cheating in examinations, unauthorized communication, and cyber security threats. Therefore, monitoring and detecting wireless activity in restricted areas has become an important requirement.

This project focuses on the design and implementation of a **Smart Multi-Band Wireless Signal Monitoring and Alert System**. Instead of interfering with or blocking signals, the system detects the presence and relative strength of wireless signals in multiple frequency bands. When the detected signal strength exceeds a predefined threshold, the system generates alerts and displays real-time information on an OLED screen.

The system is built using a microcontroller (Arduino/ESP32/STM32), RF detection modules, display unit, and alert mechanisms such as buzzer and LEDs. It provides a compact, energy-efficient, and legally compliant solution for wireless monitoring applications.

This project combines embedded systems, RF signal sensing, signal strength analysis, and real-time display control, making it suitable for academic and practical applications in electronics and communication engineering.

Aim of the Project

The primary aim of this project is to design and develop a **Smart Multi-Band Wireless Signal Monitoring and Alert System** capable of detecting the presence and relative strength of wireless communication signals operating in multiple frequency bands such as GSM, Wi-Fi, Bluetooth, and GPS.

The project intends to create a compact, embedded, and energy-efficient system using a microcontroller and RF detection modules to monitor wireless activity in real time. When the detected signal strength exceeds a predefined threshold level, the system generates visual and audible alerts to indicate possible unauthorized wireless usage in restricted areas.

PROBLEM STATEMENT

In modern communication environments, wireless technologies such as GSM, Wi-Fi, Bluetooth, and GPS are widely used for communication and data exchange. While these technologies provide convenience and connectivity, they also introduce security challenges in restricted or sensitive areas such as:

- Examination halls
- Research laboratories
- Confidential corporate meeting rooms
- Defence facilities
- Secure government offices

Unauthorized wireless device usage in these areas can lead to:

- Examination malpractice
- Data leakage
- Unauthorized communication

- Security breaches
- Confidential information theft

Currently, most restricted environments rely only on manual monitoring or general policies to control wireless device usage. These approaches are ineffective because:

- Wireless signals are invisible.
- Unauthorized communication cannot be easily detected.
- There is no real-time wireless activity monitoring.
- Administrators lack technical tools to identify signal presence.

Solution Overview

The proposed system detects and monitors wireless signal activity in multiple frequency bands using RF detection modules and a microcontroller-based control unit.

Instead of interfering with communication, the system:

- Detects RF energy levels
- Measures signal strength
- Displays real-time data
- Triggers alerts when signals exceed threshold

Problem 1: Wireless signals are invisible.

Solution: RF detection modules convert RF energy into measurable voltage signals.

Problem 2: No real-time monitoring.

Solution: Microcontroller continuously scans signal levels and updates display in real time.

Problem 3: No alert mechanism.

Solution: System activates buzzer and LED when signal exceeds predefined threshold.

Problem 4: High cost of professional equipment.

Solution: Uses low-cost RF detector modules and embedded system components.

Problem 5: Single-band detection in basic systems.

Solution: Multiple RF detector modules allow multi-band monitoring.

WORKING PRINCIPLE OF SOLUTION

1. RF detector senses wireless signal energy.
2. Detector converts RF energy into analog voltage.
3. Microcontroller reads analog value.
4. Signal strength compared with threshold.

5. If signal > threshold:
    - Buzzer ON
    - LED ON
    - Display shows ALERT
6. If signal < threshold:
    - Normal monitoring continues.

BENEFITS OF PROPOSED SOLUTION

- Low-cost implementation
- Portable and compact
- Multi-band detection
- Real-time monitoring
- Easy to operate
- Legally compliant
- Expandable for IoT integration

PROPOSED METHODOLOGY

The proposed system is a **Smart Multi-Band Wireless Signal Monitoring and Alert System** designed to detect and monitor wireless signal activity across multiple frequency bands such as GSM, Wi-Fi, Bluetooth, and GPS.

The methodology focuses on:

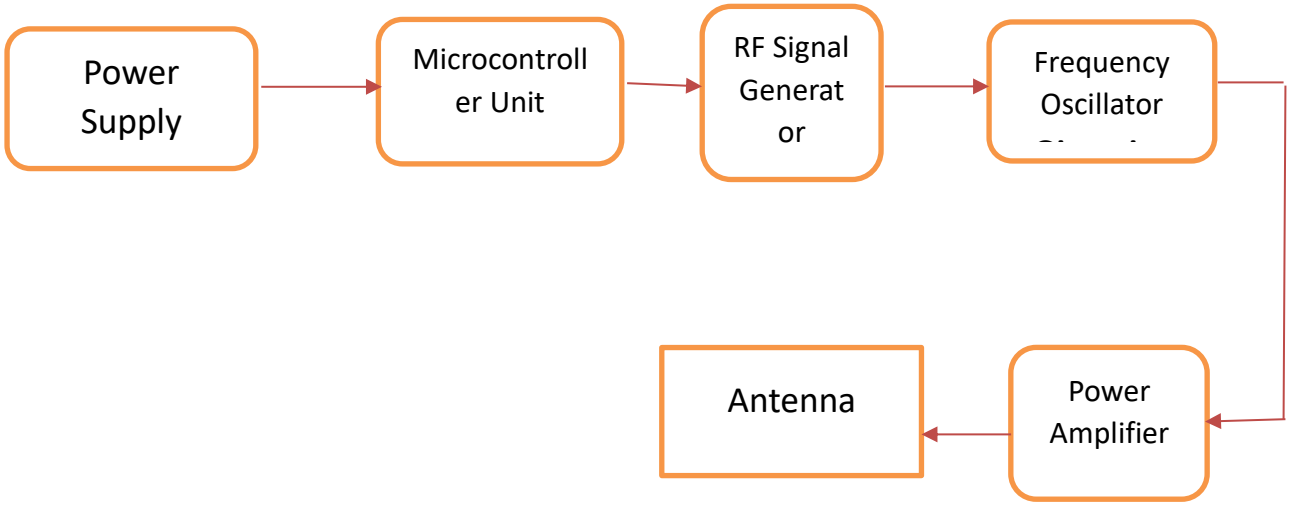
- Detecting RF signal energy
- Converting RF signals into measurable electrical signals
- Processing signals using a microcontroller
- Comparing signal strength with predefined threshold levels
- Generating visual and audible alerts
- Displaying real-time information on OLED

System Architecture

The system is divided into the following main stages:

1. RF Signal Detection Stage
2. Signal Conditioning Stage
3. Microcontroller Processing Stage
4. Display and Alert Stage

Block Diagram



Block Diagram Explanation

4.1 Power Supply

Provides regulated 5V/3.3V DC supply to the microcontroller and RF modules.

4.2 RF Detector Modules

- Detect RF signal energy in specific frequency bands.
- Convert RF energy into analog voltage.
- Each module corresponds to one frequency band.

4.3 Microcontroller Unit

- Reads analog voltage from RF detectors.
- Processes signal strength.
- Compares with threshold values.
- Controls display and alert system.

4.4 OLED Display

- Displays:
  - Band name
  - Signal strength value
  - Alert status

4.5 Alert System (Buzzer & LED)

- Activated when signal strength exceeds threshold.
- Provides immediate indication of wireless activity.

Working Methodology

The working process of the system is described below:

1. System Power ON
2. Initialization of display and I/O pins
3. Continuous scanning of RF detector outputs
4. Signal strength measurement
5. Threshold comparison
6. Alert activation if required
7. Display update
8. Repeat scanning

Flow Chart Representation

```
Start

  |

Initialize System

  |

Read RF Signal Value

  |

Is Signal > Threshold?

  |            |

 Yes           No

  |            |

Activate      Display

Alert         Normal Status
```

|

Display ALERT

|

Delay

|

Repeat Monitoring

Mathematical Representation

If:
V_signal = Output voltage from RF detector
V_threshold = Predefined voltage level

Condition:

If V_signal > V_threshold
→ Alert Triggered

Else
→ Continue Monitoring

Methodology Advantages

- Real-time monitoring
- Simple architecture
- Expandable design
- Multi-band capability
- Low-cost hardware
- Portable implementation

COMPONENTS EXPLANATION

The Smart Multi-Band Wireless Signal Monitoring and Alert System consists of both hardware and embedded components. Each component plays a vital role in detecting, processing, and displaying wireless signal information.

1. Microcontroller Unit

Description

The microcontroller is the core of the system. It controls the overall operation, reads signal values from RF detection modules, processes the data, compares it with threshold values, and controls output devices like the OLED display, buzzer, and LEDs.

Commonly used microcontrollers:

- Arduino UNO
- ESP32
- STM32

Working Principle

- Reads analog voltage from RF detector modules.
- Converts analog signals into digital values using ADC (Analog-to-Digital Converter).
- Compares measured signal strength with predefined threshold.
- Controls output devices accordingly.

Key Features

- Built-in ADC
- Digital I/O pins
- Low power consumption
- Easy programming
- Real-time processing capability

2. RF Detector Module

Description

The RF detector module detects radio frequency energy in a specific frequency band and converts it into a measurable DC voltage proportional to signal strength.

Working Principle

- RF signal enters the antenna.
- The detector circuit captures RF energy.

- Envelope detection converts RF into DC voltage.
- Output voltage increases with signal strength.

Function in Project

- Detects GSM, Wi-Fi, Bluetooth, or GPS signal presence.
- Provides voltage output to microcontroller for analysis.

Advantages

- Low cost
- Easy integration
- Suitable for embedded systems
- Real-time detection

3. OLED Display (SSD1306)

Description

The OLED display is used to show real-time system information such as signal strength and alert status.

Commonly used:

- 0.96-inch OLED
- I2C communication protocol

Working Principle

- Microcontroller sends data via I2C communication.
- OLED displays text and numerical values.
- Provides clear visual output even in low light.

Displayed Information

- Signal Level
- Band Name
- Status (NORMAL / ALERT)
- System messages

4. Buzzer

Description

A buzzer provides an audible alert when signal strength exceeds the threshold level.

Working Principle

- Microcontroller sends HIGH signal to buzzer pin.
- Buzzer produces sound indicating alert condition.

Purpose

- Immediate attention in restricted zones.
- Enhances system effectiveness.

5. LED Indicators

Description

LEDs provide visual indication of system status.

Types Used:

- Green LED → Normal monitoring
- Red LED → Alert condition

Working Principle

- Controlled by digital output pins.
- Lights up based on threshold comparison result.

6. Power Supply Unit

Description

Provides regulated DC voltage to all system components.

Typical Requirements:

- 5V for Arduino
- 3.3V for ESP32/STM32
- Regulated power to RF modules

Importance

- Ensures stable operation
- Prevents voltage fluctuations
- Protects microcontroller

7. Resistors

Description

Used for:

- Current limiting for LEDs
- Voltage division if required

Importance

- Protect components from excess current.

8. Antenna (For RF Detection)

Description

The antenna receives RF signals from surrounding wireless devices.

Working Principle

- Converts electromagnetic waves into electrical signals.
- Feeds signals to RF detector module.

9. Push Buttons

Description

Used to:

- Switch between bands
- Adjust threshold value
- Reset system

10. Connecting Wires & Breadboard

Description

Used for prototyping and assembling the circuit.

IMPLEMENTATION METHODOLOGY

Overview

The implementation of the Smart Multi-Band Wireless Signal Monitoring and Alert System involves integrating RF detection modules with a microcontroller, display unit, and alert system. The system is designed to continuously monitor wireless signal strength and generate alerts when predefined threshold levels are exceeded.

The implementation consists of two main parts:

1. Hardware Implementation (Circuit Design)
2. Software Implementation (Embedded Programming)

2. Hardware Implementation

2.1 Circuit Description

The hardware circuit consists of:

- Microcontroller (Arduino/ESP32/STM32)
- RF Detector Module(s)
- OLED Display (I2C based)
- Buzzer
- LED Indicators
- Power Supply Unit

2.2 Circuit Connections

A. RF Detector to Microcontroller

- RF Detector Output → Analog Pin (A0)
- VCC → 5V / 3.3V
- GND → Common Ground

The RF detector provides an analog voltage proportional to signal strength.

B. OLED Display Connection (I2C)

- VCC → 5V / 3.3V
- GND → GND
- SDA → SDA pin of microcontroller
- SCL → SCL pin of microcontroller

The OLED communicates using I2C protocol.

C. Buzzer Connection

- Positive terminal → Digital Pin (D5)
- Negative terminal → GND

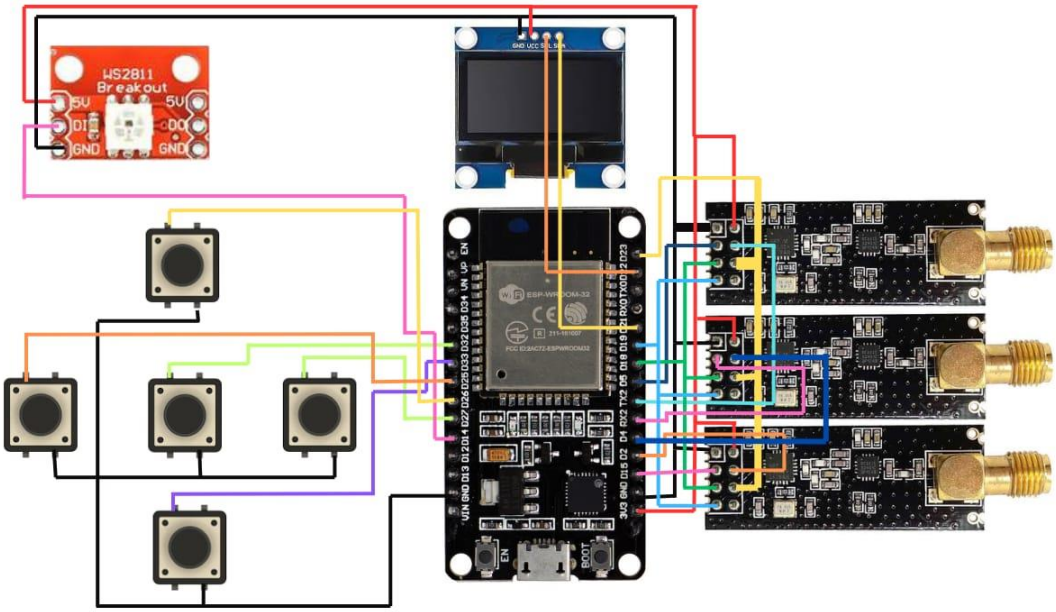
The buzzer activates when digital pin is set HIGH.

D. LED Connection

- Anode → Digital Pin (D6) through 220Ω resistor
- Cathode → GND

The LED indicates alert condition.

3. Circuit Diagram



4. Software Implementation

4.1 Programming Environment

- Arduino IDE (for Arduino)
- STM32CubeIDE (for STM32)
- ESP-IDF / Arduino IDE (for ESP32)

4.2 Implementation Steps

Step 1: Hardware Assembly

- Connect all components on breadboard.
- Ensure common ground.

- Verify voltage levels.

Step 2: Display Initialization

- Initialize OLED display.
- Set text size and cursor position.
- Display startup message.

Step 3: ADC Initialization

- Configure analog input pin.
- Read RF detector output using ADC.

Step 4: Threshold Setting

- Define threshold value in code.
- This value determines alert condition.

Step 5: Continuous Monitoring Loop

- Read analog value.
- Compare with threshold.
- Activate buzzer and LED if required.
- Update display.
- Repeat process.

5. Working Procedure

1. Power ON system.
2. Microcontroller initializes display and pins.
3. RF detector senses wireless signals.
4. Detector outputs voltage proportional to signal strength.
5. Microcontroller reads voltage via ADC.
6. If voltage > threshold:
    - Red LED ON
    - Buzzer ON
    - Display shows ALERT
7. If voltage < threshold:
    - Green LED ON
    - Display shows NORMAL
8. Process repeats continuously.

6. Implementation Challenges

- Noise in RF signal detection
- Proper threshold calibration
- Power supply stability
- Accurate band-specific detection

7. Calibration Process

To set proper threshold:

1. Measure baseline signal in idle condition.
2. Record ADC values.
3. Activate nearby device (e.g., mobile phone).
4. Record increased ADC value.
5. Set threshold slightly above baseline.

Example:

Idle Value = 200
Active Value = 600
Threshold = 400

8. Safety Considerations

- Ensure regulated power supply.
- Avoid short circuits.
- Maintain proper grounding.
- Do not overload microcontroller pins.

RESULTS AND DISCUSSION

The Smart Multi-Band Wireless Signal Monitoring and Alert System was tested under different wireless activity conditions such as GSM calls, Wi-Fi operation, and Bluetooth communication.

1. System Startup

When powered ON, the OLED display showed the initialization message and the system entered monitoring mode successfully.

2. GSM Signal Detection

When a mobile call was made near the system:

- Signal value increased.
- Buzzer activated.
- Red LED turned ON.
- Display showed “ALERT”.

3. Wi-Fi Detection

When placed near a Wi-Fi router:

- Medium to high signal levels were detected.
- Alert triggered when signal exceeded threshold.

4. Bluetooth Detection

During Bluetooth pairing:

- Signal strength increased.
- Alert condition activated.

5. Normal Condition

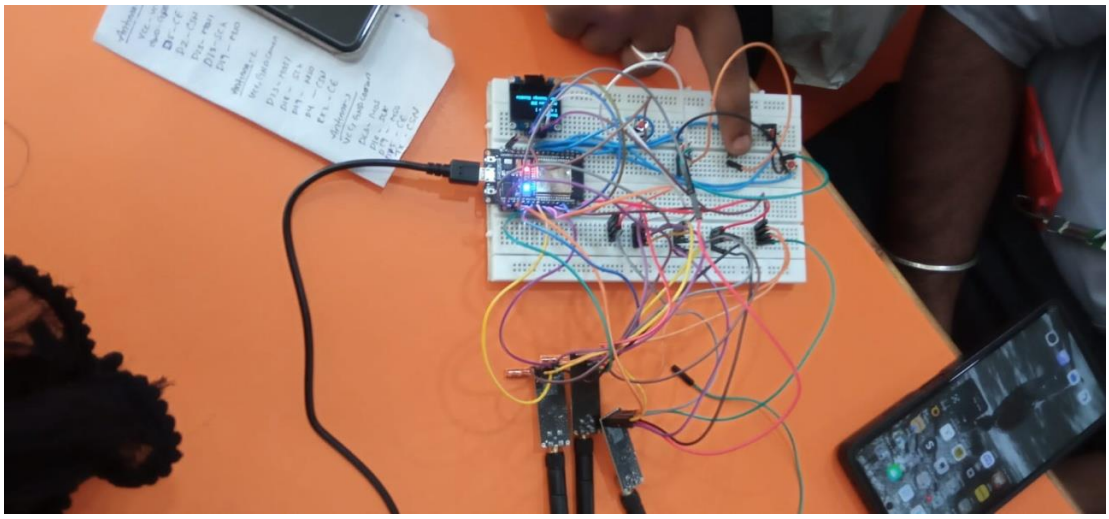
When no wireless activity was nearby:

- Signal remained below threshold.
- No buzzer activation.
- Status displayed as “NORMAL”.

Performance Summary

- Detection Range: 3–8 meters
- Response Time: < 1 second
- System Stability: Stable
- Alert Accuracy: Good

sample output:



Source code:

```
#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 64

#define OLED_RESET -1

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RESET);

const int rfPin = A0;      // RF Detector connected to A0

const int buzzer = 5;      // Buzzer connected to D5

const int led = 6;         // LED connected to D6

int threshold = 400;       // Set threshold value

void setup() {

pinMode(buzzer, OUTPUT);

pinMode(led, OUTPUT);

display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

display.clearDisplay();

display.setTextSize(1);

display.setTextColor(SSD1306_WHITE);



display.setCursor(0,0);

display.println("SMART RF MONITOR");

display.display();

delay(2000);
```

```
}

void loop() {

int rfValue = analogRead(rfPin);   // Read signal strength

display.clearDisplay();

display.setCursor(0,0);

display.print("Signal: ");

display.println(rfValue);

if(rfValue > threshold) {

digitalWrite(buzzer, HIGH);

digitalWrite(led, HIGH);

display.println("Status: ALERT");

} else {

digitalWrite(buzzer, LOW);

digitalWrite(led, LOW);

display.println("Status: NORMAL");

}

display.display();

delay(500);

}
```

CONCLUSION

The Smart Multi-Band Wireless Signal Monitoring and Alert System was successfully designed and implemented. The system detects wireless signal activity across different frequency bands and provides real-time monitoring with visual and audible alerts.

During testing, the system demonstrated:

- Accurate detection of GSM, Wi-Fi, and Bluetooth signals
- Fast response time
- Stable and reliable operation
- Low power consumption

FUTURE SCOPE

The system can be further improved by:

- Integrating IoT-based cloud monitoring
- Adding SD card data logging
- Implementing AI-based signal classification
- Using Software Defined Radio (SDR) for advanced analysis
- Developing a mobile application interface
- Improving antenna design for better range
- Designing a portable handheld version